

1. Let $f : D \rightarrow \mathbb{R}$ be a function.
 - (a) Prove that if f is continuous at $x_0 \in D$, then $|f|$ is continuous at x_0 .
 - (b) Is the converse true? Prove it or give a counterexample.
2. Show that every function $f : \mathbb{N} \rightarrow \mathbb{R}$ is uniformly continuous.
3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, and assume that there exists some constant $0 < C < 1$ such that

$$|f(x) - f(y)| \leq C|x - y|$$

for all $x, y \in \mathbb{R}$.

- (a) Prove that f is continuous on $\mathbb{R}$.
- (b) Let $a_0 = 2023$ and $a_n = f(a_{n-1})$. Prove that this sequence converges by showing it is Cauchy.
- (c) Let $A = \lim_{n \rightarrow \infty} a_n$. Prove that A is a fixed point of f , i.e., $f(A) = A$.
- (d) Prove that this A is the only fixed point of f , i.e., if $B \in \mathbb{R}$ satisfies $f(B) = B$, then $B = A$.